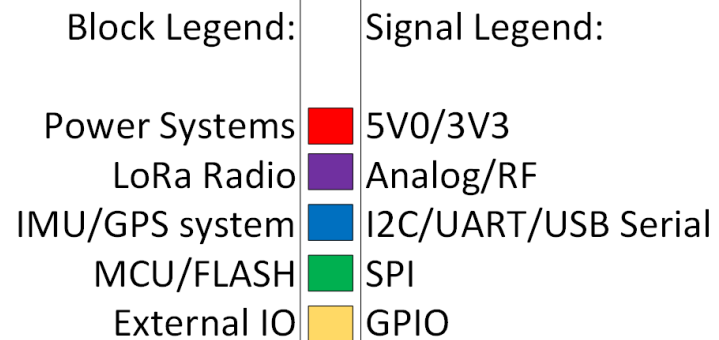
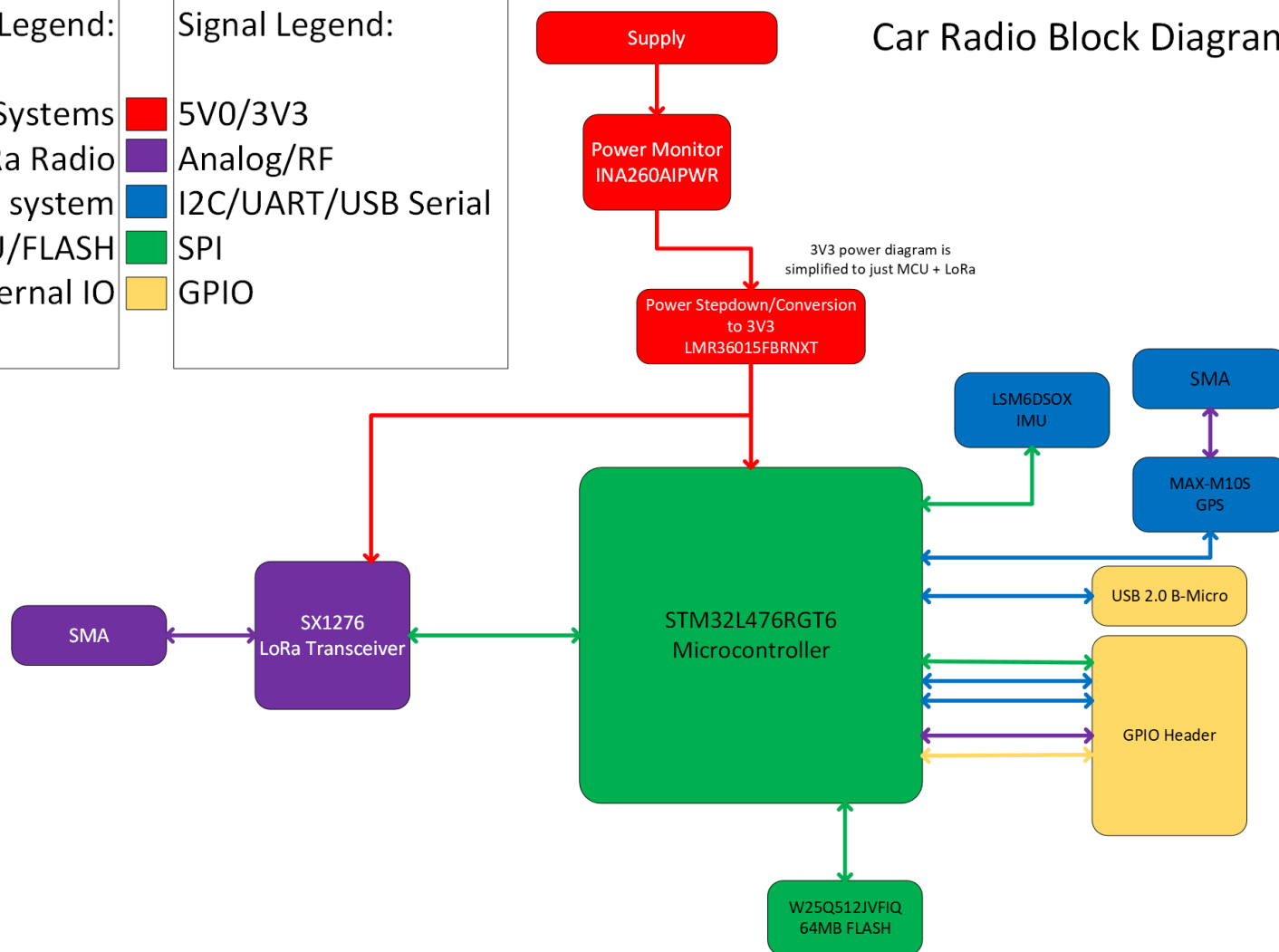


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questions																						
A	Table of Contents 1. Title Sheet and TOC 2. Block Diagram 3. MCU 4. Flash Memory 5. LoRa Transceiver 6. Power Monitor 9. Power Regulator 10. Header 11. USB 12. GPS 13. IMU						A															
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C							C															
D	<table><tr><td colspan="3">Title</td></tr><tr><td>Size</td><td>Number</td><td>Revision</td></tr><tr><td>A</td><td></td><td></td></tr><tr><td>Date:</td><td>4/03/2025</td><td>Sheet of</td></tr><tr><td>File:</td><td>C:\Users\...\Top.SchDoc</td><td>Drawn By:</td></tr></table>						Title			Size	Number	Revision	A			Date:	4/03/2025	Sheet of	File:	C:\Users\...\Top.SchDoc	Drawn By:	D
Title																						
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Car Radio Block Diagram:

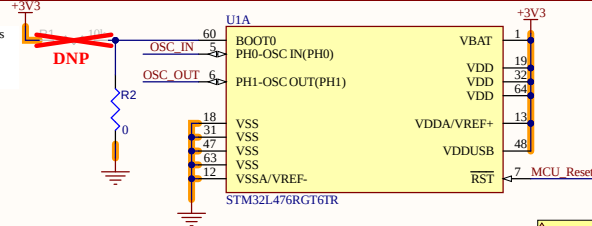


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File:	C:\Users\...\Block_Diagram.SchDoc	Drawn By:

STM32 Microcontroller - STM32L476RGT6

MCU Power Circuitry

The "DNP" underneath [R1] is a temporary fix, until we can get Altium to output PDFs with DNP markers on them

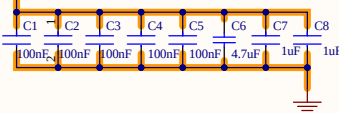


U1A: VDD, VBAT, VDDA/VREF+, VDDUSB use +3V3
Page 109 of DS lists 100nF cap/supply pin

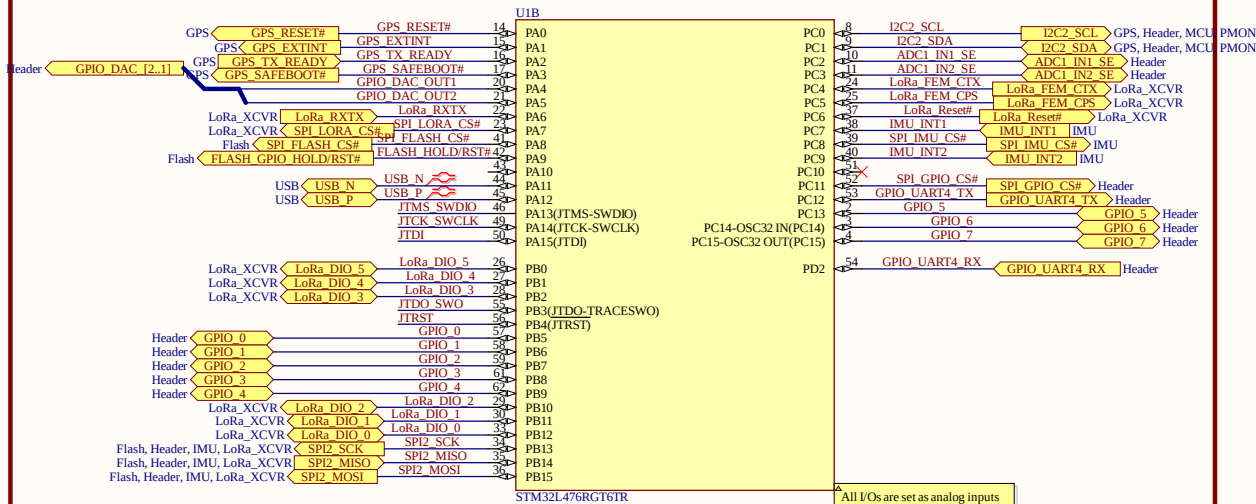
Decoupling Capacitors for STM32. DS requires 1 DEC per port.

U1A: BOOT0 pin has three options. 0 (GND) correlates to Main Flash Boot and executes user code from Flash memory (default mode). 1 (VDD) w/ nBOOT1 Option Bit: 0 correlates to System Memory Bootloader and allows for firmware updates. 1 (VDD) w/ nBoot1 Option Bit: 1 correlates to SRAM Boot, which Boots into RAM for debugging.

VDD 4.7uF bulk
VDDA,VDDUSB 2X1uF
Updated: VBAT, 3xVDD, 1xVDDA100nF

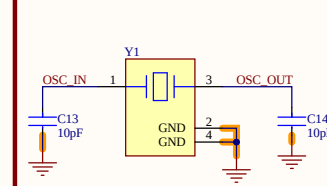


MCU

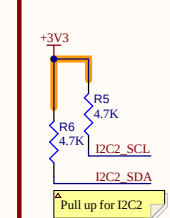


All I/Os are set as analog inputs after reset. Need Pull up/down for unused pins if external interference is an issue.

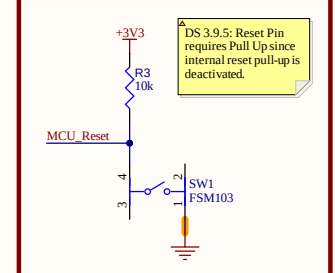
16 MHz Crystal



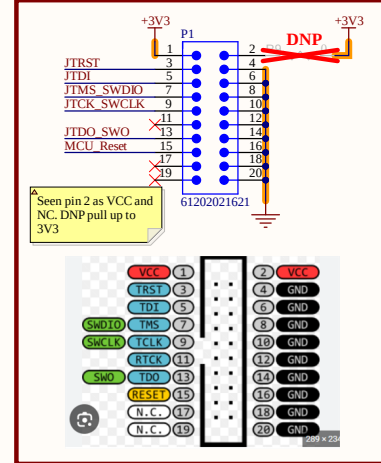
I2C Pull Up



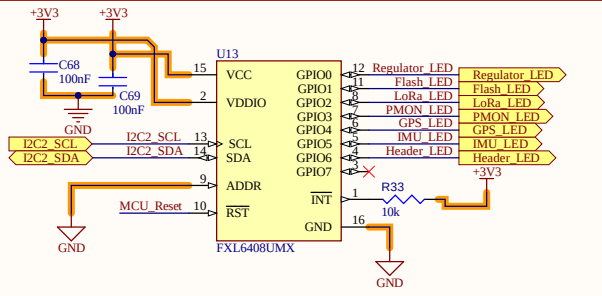
MCU Reset Pull-Up



JTAG Debugger



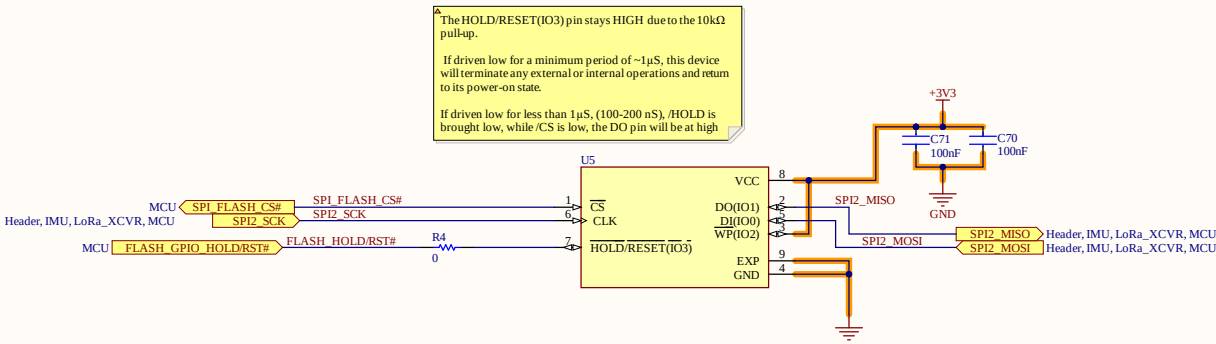
GPIO Extender



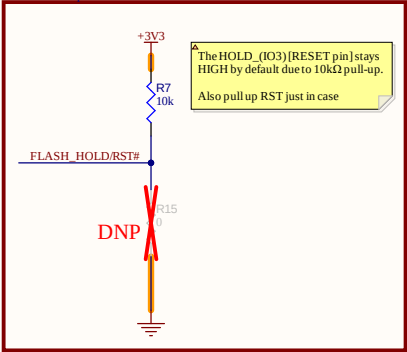
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30 51,52,53 5457
2,3,4? 61,62
8,9

FLASH Memory - W25Q512JVFIQ



FLASH Pull-Up Resistor

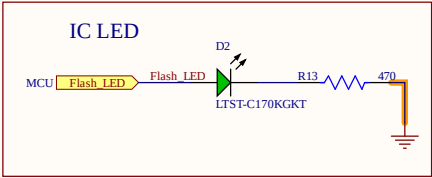


The RESET pin is a dedicated hardware reset pin regardless of device settings or operation states. If the hardware reset function is not used, this pin can be left floating or connected to VCC in the system.

Pin 1 (/HOLD_(IO3)) has been dedicated as reset pin.

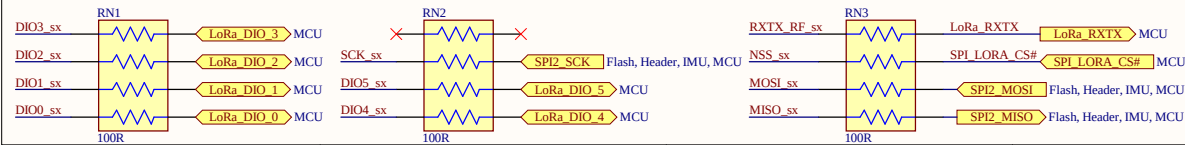
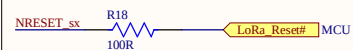
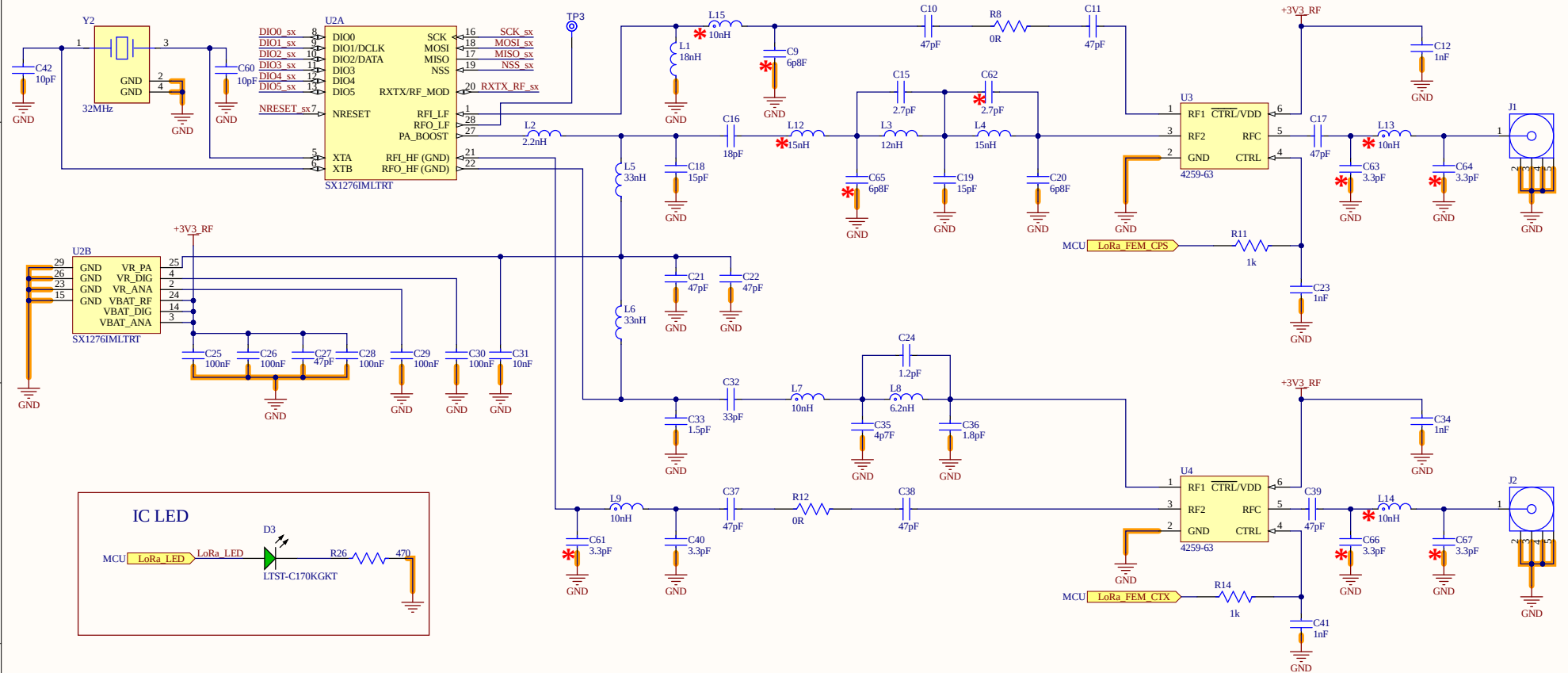
WP_IO2 is tied to 3V3 to disable write protection, ensuring unrestricted read and write operations to flash memory.

IC LED



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A3		
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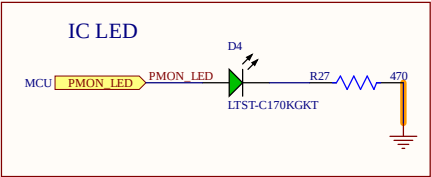
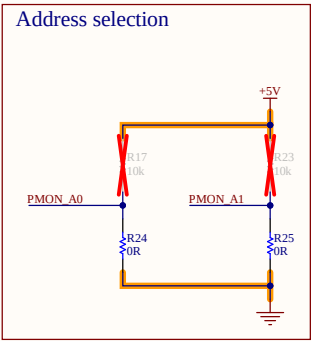
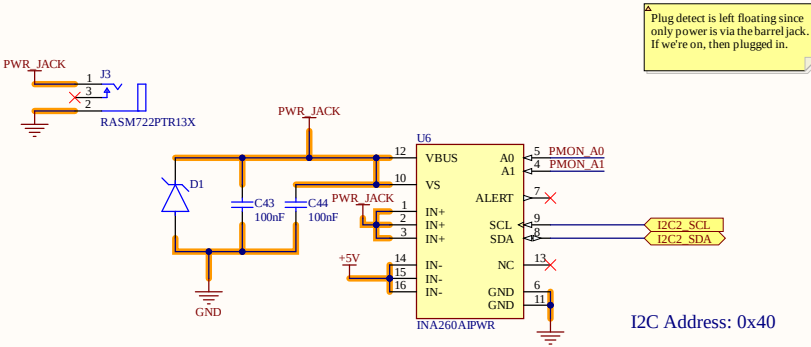
LoRa Transciever - SX1276



GPIO = [LoRa_DIO_0...LoRa_DIO_5]
 NetLabels and Ports
 SPI =
 NSS --> LoRa_CS (NetLabel and Port)
 MOSI = SPI2_MOSI
 MISO = SPI2_MISO
 SCK = SPI2_SCK (Both nets and ports)
 RXTX = LoRa_Reset#
 I2C2_SCL and I2C2_SDA
 LoRa_FEM_CTX
 LoRa_FEM_CPS

Title		
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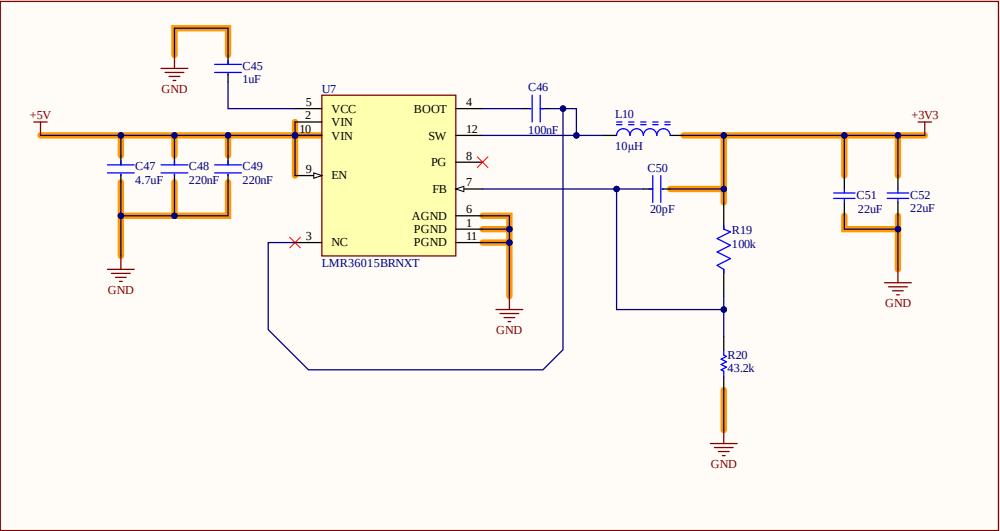
Power Monitor - INA260AIPWR



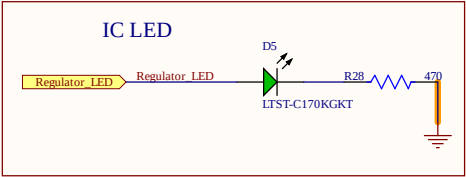
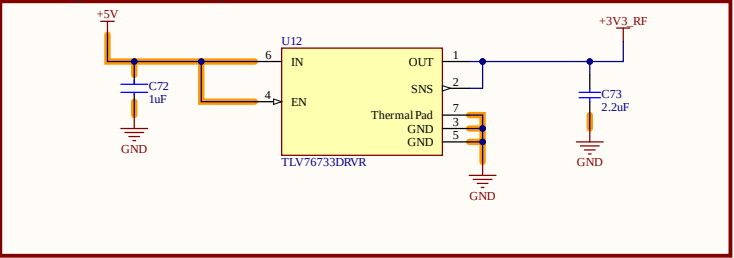
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Power Regulator

Buck Converter



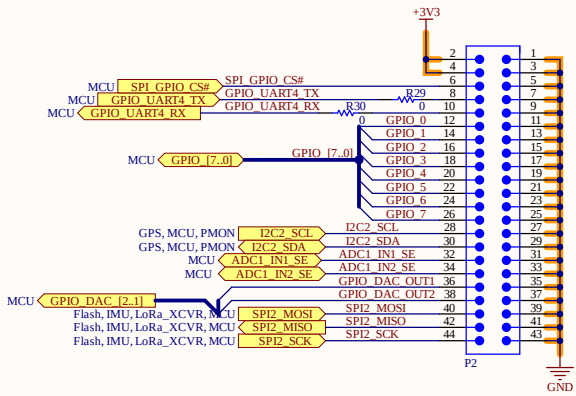
Linear Regulator For LoRa Chip



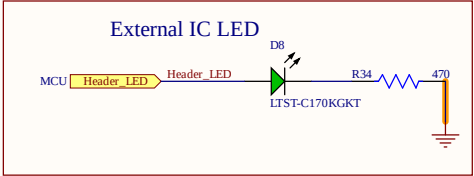
Title		
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A3		
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File:	C:\Users\...\Regulator.SchDoc	Drawn By:

GPIO Header

8 X GPIO, 2 x ADC, 2 x DAC (have to find dac), 2- UART: (TX/RX), 2- I2C: (SDA/SCL), 4 - SPI: (MOSI/MISO/SCK/CS), 2 X +3V3. All need ground, so 44 total pins.

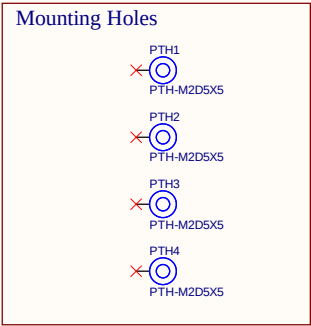
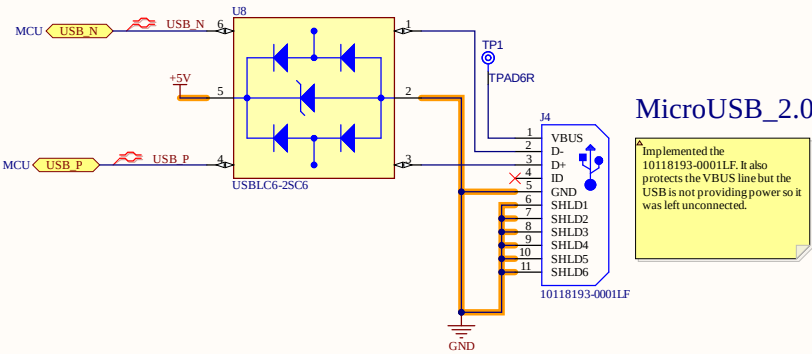


gpio7..0
12c3



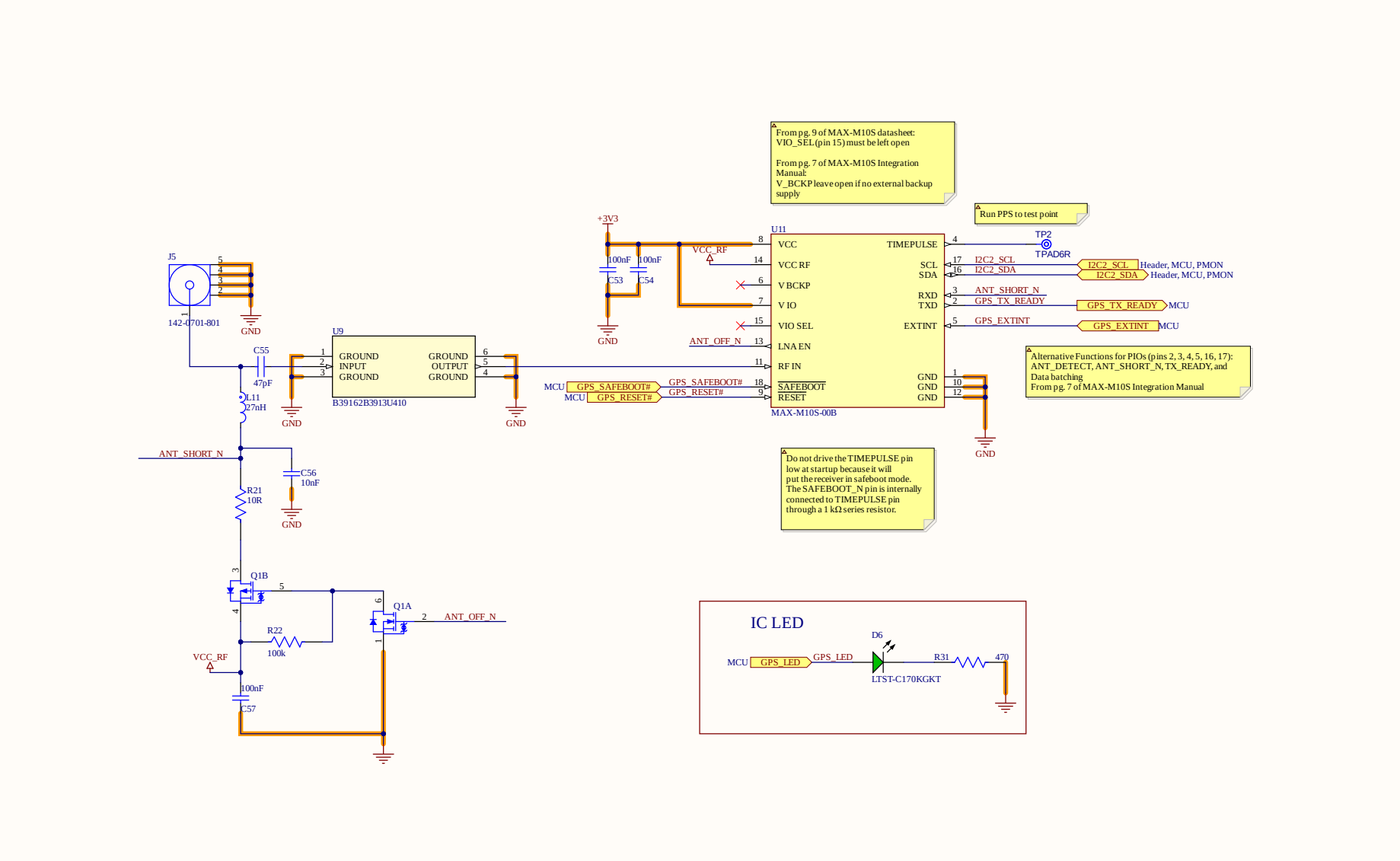
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USB



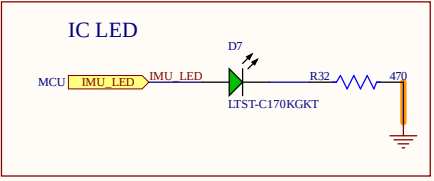
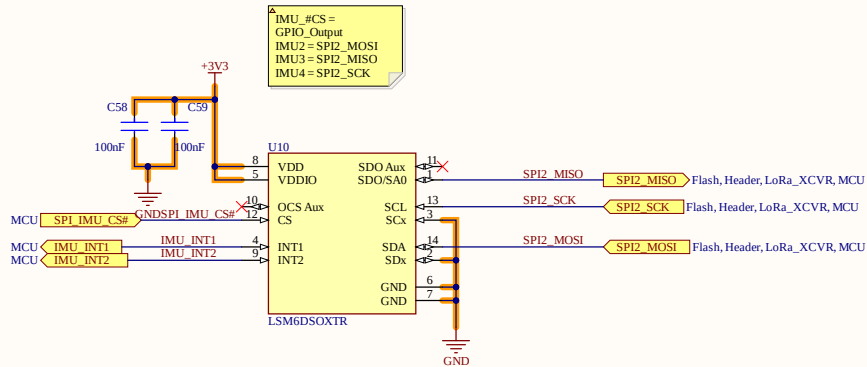
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IMU Module



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